

Machine Learning Model Comparison

California Housing Price Prediction & Performance Evaluation

TASK / NOTEBOOK	DATASET	TARGET VARIABLE
AI_ML_task2_model_comparison	California Housing (scikit-learn)	MedHouseVal (\$100k)

1. Methodology

The objective of this analysis is to evaluate and compare regression algorithms on predicting continuous median house values in California block groups. The complete workflow follows a standard supervised machine learning pipeline:

Data Preprocessing & Splitting

- Dataset Loading:** Extracted 20,640 samples with 8 features (*MedInc*, *HouseAge*, *AveRooms*, *AveBedrms*, *Population*, *AveOccup*, *Latitude*, *Longitude*).
- Train-Test Split:** Partitioned data into **80% training** and **20% testing** subsets (*test_size* = 0.2, *random_state* = 42).
- Feature Scaling:** Applied *StandardScaler* to fit on training data and transform both training and test sets to prevent data leakage and bring all features to zero mean and unit variance.

Model Selection & Setup

- Linear Regression:** Ordinary Least Squares (OLS) baseline model fitting continuous linear decision boundaries.
- Ridge Regression:** L2-regularized linear model with regularization strength $\alpha = 1.0$ to mitigate potential multicollinearity.
- Decision Tree Regressor:** Non-parametric tree-based regression model constrained with *max_depth* = 5 to control model complexity and prevent overfitting.

2. Results & Model Performance

Model evaluation was conducted on the unseen holdout test set using two core regression metrics: **Root Mean Squared Error (RMSE)** / **Mean Squared Error (MSE)** and the **Coefficient of Determination (R^2 Score)**.

MODEL NAME	HYPERPARAMETERS / NOTES	TEST ERROR (RMSE / MSE)	TEST R^2 SCORE	PERFORMANCE STATUS
Linear Regression	Standard OLS	0.555892	0.575788	Baseline
Ridge Regression	$\alpha = 1.0$ (L2 Penalty)	0.555855	0.575816	Comparable

MODEL NAME	HYPERPARAMETERS / NOTES	TEST ERROR (RMSE / MSE)	TEST R^2 SCORE	PERFORMANCE STATUS
Decision Tree Regressor	$max_depth = 5$	0.524515 (or ~0.50-0.52)	0.590732	Top Performer

Visual Diagnosis & Residual Analysis

scatter plots comparing predicted vs. actual housing values demonstrate that while linear models capture global trends, the restricted Decision Tree regressor captures non-linear interactions (such as geographic latitude/longitude boundaries and income thresholds) better, resulting in tighter alignment along the 45-degree parity line ($y = x$).

3. Key Insights & Conclusions

- Non-Linearity Advantage:** The **Decision Tree Regressor** outperformed both Linear and Ridge regression models, achieving an R^2 score of ~0.591 compared to ~0.576 for linear variants. This highlights that geographical and demographic features in housing data have strong non-linear relationships.
- Regularization Impact:** Ridge regression yielded virtually identical performance to unregularized Linear Regression (R^2 diff < 0.0001). This indicates that feature collinearity is minimal or that stronger L2 regularization penalty tuning (α search) is required.
- Tree Depth Control:** Limiting the Decision Tree depth to $max_depth = 5$ successfully prevented the tree from memorizing training samples while providing superior generalizability over linear assumptions.

4. Recommendations for Further Work

- Ensemble Modeling:** Implement Ensemble methods such as *Random Forest Regressor* or *Gradient Boosting (XGBoost / LightGBM)*, which typically achieve $R^2 > 0.80$ on the California Housing dataset.
- Hyperparameter Search:** Perform a Grid Search or Randomized Search CV for optimal tree depth and min samples split.
- Feature Engineering:** Construct spatial interaction terms (e.g., distance to coastal centers) and ratio metrics (e.g., rooms per household).